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# The Empirical Performance of RSI-Based Trading Strategies

*A Backtest Across 99 U.S. Equities on Real Prices (1962–2026)*

Author John, CaesarBot & luppiter Group

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ABSTRACT

This study backtests four Relative Strength Index (RSI) trading strategies on real daily prices for 99 large U.S. stocks in 11 sectors, using each stock's full Yahoo Finance history from as early as January 1962 through October 30, 2026. The strategies buy when the 14-day RSI crosses below 30 or 20 and sell when it crosses above 70 or 80, on daily or weekly data, and every trade is executed at the next day's opening price on dividend-adjusted prices. The rerun reproduces the original paper closely: the Daily 30/70 rule made 3,948 trades with a 79.9% win rate and a 7.63% average return, against 3,892, 79.4%, and 7.66% originally.

> The new tests change how those results should be read. Against random entries on the same stocks held for the same lengths of time, RSI entries won far more often (79.9% against 64.3% for Daily 30/70), but their average return was only modestly higher (7.63% against 6.96%, p = 0.032), and for Daily 20/80 it was no higher at all; much of the high win rate comes from the exit rule, which sells only after a rally. Stock by stock, even with Treasury bill interest on idle cash, every strategy trailed buying and holding the same stock (Daily 30/70: 8.2% a year against 13.7%, ahead in only 11 of 99 stocks), because capital sat out of the market about 62% to 88% of the time. As equal-weight portfolios from 1988, the strategies returned 5.5% to 10.3% a year with much lower volatility than the S&P 500 and higher Sharpe ratios than the index (0.65 to 0.80 against 0.51), but below the index's 11.4% return and well below the 18.1% earned by simply holding the same stocks. Because the universe is made of companies that survived to 2026, and holding them returned far more than the index, all absolute results are biased upward; the comparison with holding the same stocks is the fair one, and on that comparison the rules lost.

Data: Yahoo Finance daily prices, retrieved November 10, 2026. All figures are computed from actual historical prices.

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SECTION I

# 1. Introduction

The Relative Strength Index (RSI), introduced by J. Welles Wilder in his seminal 1978 work New Concepts in Technical Trading Systems, is among the most widely used momentum oscillators in technical analysis (Wilder, 1978). The RSI measures the magnitude of recent price changes on a scale of 0 to 100, with readings below 30 traditionally interpreted as "oversold" and readings above 70 as "overbought." Technical analysis is widely used by practitioners: in a Bank of England survey of chief foreign exchange dealers in London, about 90% of respondents said they used some chart analysis in forming exchange rate expectations at short horizons (Taylor & Allen, 1992). Even so, the academic literature on RSI's profitability remains surprisingly thin relative to studies on moving average crossover systems.

The efficient market hypothesis (EMH), as formulated by Fama (1970), posits that asset prices fully reflect all available information, implying that technical trading rules should not generate systematic excess returns. If markets are efficient, past price patterns, the sole input to RSI, should have no predictive power for future returns. Yet behavioral finance research has documented systematic overreaction in stock prices (De Bondt & Thaler, 1985), providing a theoretical basis for why mean-reversion indicators like RSI might capture genuine profit opportunities.

This study makes three contributions to the literature. First, we employ a substantially larger stock universe (99 equities across 11 sectors) and longer time period (up to 64 years) than prior RSI backtests, which typically examine a handful of indices or a single market. Second, we systematically compare four threshold/timeframe combinations rather than testing a single parameterization; reporting all four limits selective reporting, though it does not by itself resolve the data-snooping concerns raised by Sullivan, Timmermann, and White (1999) (see Section 5.7). Third, we test the trades against random entries on the same stocks, against the S&P 500 over the same dates, and against buying and holding the same stocks, both stock by stock and as portfolios with Treasury bill interest on idle cash, and we break results down by sector, decade, and crisis window.

SECTION II

# 2. Literature Review

## 2.1 Technical Analysis and Market Efficiency

The debate over technical analysis profitability stretches back decades. Fama (1970) established the efficient markets framework that implied technical trading rules should not work. However, Brock, Lakonishok, and LeBaron (1992) challenged this view with evidence that simple moving average and trading range breakout rules generated statistically significant returns on the Dow Jones Industrial Average from 1897 to 1986; those returns are reported before transaction costs. Their findings sparked renewed academic interest in technical analysis.

Lo, Mamaysky, and Wang (2000) provided rigorous statistical foundations for technical analysis by formalizing pattern recognition algorithms and demonstrating that several technical patterns provided incremental information beyond what was captured by standard parametric models. Park and Irwin (2007), in their comprehensive survey of 95 modern studies, found that 56 reported positive results for technical trading rules, 20 reported negative results, and 19 were mixed, suggesting that the evidence is tilted toward, but not conclusive about, profitability. Malkiel (2003), surveying the behavioral and statistical challenges to the efficient market hypothesis, concluded that stock markets are more efficient and less predictable than many academic papers of the time suggested.

More recently, Neely, Rapach, Tu, and Zhou (2014) demonstrated that technical indicators have significant out-of-sample predictive power for the equity risk premium, complementing traditional macroeconomic predictors. Urquhart and McGroarty (2016) tested return predictability in the S&P 500, FTSE 100, Nikkei 225, and EURO STOXX 50 from 1990 to 2014 and found periods of significant predictability alternating with periods of none, consistent with the Adaptive Markets Hypothesis (Lo, 2004); their tests measure statistical predictability rather than the profitability of technical trading rules.

## 2.2 RSI-Specific Research

Research specifically focused on the RSI is more limited. Chong and Ng (2008) tested RSI and MACD rules on 60 years of data for the London Stock Exchange's FT30 index and found that both could generate returns higher than buy-and-hold in most cases. Wong, Manzur, and Chew (2003) tested the moving average, a trend-following indicator, and the RSI, a counter-trend indicator, on Singapore data and found that both could generate significantly positive returns.

Chong, Ng, and Liew (2014) revisited the FT30 findings on five other OECD stock markets and found that RSI profitability depended on both the market and the parameterization: an RSI(21, 50) rule generated significant abnormal returns on the Milan Comit General and S&P/TSX Composite indices, while the conventional RSI(14) 30/70 rule was profitable on the Dow Jones Industrial Average. Metghalchi, Chen, and Hayes (2015) tested moving average, RSI, MACD, and trading breakout rules on the Madrid General Stock Index from 1975 to 2012 and reported predictive power that was statistically significant in all three of their sub-periods, along with strategies that may outperform buy-and-hold after transaction costs and risk. Like Chong and Ng (2008), these index-level results cut against this paper's finding that every rule lagged buy-and-hold stock by stock; the studies differ in market, period, and rule design, and this paper does not test which difference accounts for the gap. Bajgrowicz and Scaillet (2012), applying a false discovery rate approach to daily Dow Jones Industrial Average data from 1897 to 2011, found that investors could not have reliably selected the best rules in advance and that in-sample profits disappear once even low transaction costs are included.

Han, Yang, and Zhou (2013) applied a moving average timing rule to portfolios sorted by volatility and found that the timed portfolios beat buy-and-hold, with abnormal returns for high-volatility portfolios that the CAPM and the Fama-French three-factor model do not explain. Their study tests moving averages, not the RSI, but it documents cross-sectional differences in the payoff to a technical rule.

### 2.3 Behavioral Foundations

The potential profitability of RSI-based strategies finds theoretical support in behavioral finance. De Bondt and Thaler (1985) documented long-term return reversals consistent with investor overreaction: prior losers outperformed prior winners over 3–5 year horizons. Over shorter horizons the pattern runs the other way: Jegadeesh and Titman (1993) found that buying past winners and selling past losers earned significant positive returns over 3- to 12-month holding periods, a range that includes the typical Daily 30/70 holding period in this study. Tversky and Kahneman (1974) identified representativeness bias and anchoring as cognitive heuristics that can cause systematic pricing errors. Shiller (2000) argued that irrational exuberance and panic create price overshoots in both directions, which mean-reversion indicators might exploit.

If stocks that trigger RSI oversold signals have indeed been subject to excessive selling pressure driven by behavioral biases, then the subsequent rebound captured by a buy-on-oversold/sell-on-overbought strategy reflects a correction of mispricing rather than compensation for risk.

SECTION III

## 3. Data and Methodology

### 3.1 Stock Universe and Data

The universe is the original paper's list of 100 stocks, chosen as large companies listed in 2026. One of them can no longer be tested: AvalonBay Communities merged into Equity Residential in August 2026, and Yahoo Finance no longer serves AvalonBay's price history, so the rerun covers 99 stocks. Equity Residential continues under the ticker VMRK and keeps its full history. The loss of AvalonBay is a small, live example of the survivorship problem discussed in Section 5.5.

Table A: Stock Universe

| Sector | Stocks | Tickers |
|---|---|---|
| Mega Cap Tech | 7 | AAPL, MSFT, GOOGL, AMZN, NVDA, META, TSLA |
| Financials | 10 | JPM, BAC, GS, WFC, MS, BRK-B, C, AXP, BLK, SCHW |
| Healthcare | 10 | JNJ, UNH, PFE, ABBV, MRK, LLY, TMO, ABT, BMY, AMGN |
| Energy | 10 | XOM, CVX, COP, SLB, EOG, MPC, PSX, VLO, OXY, HAL |
| Consumer Staples | 10 | PG, KO, PEP, WMT, COST, PM, MO, CL, GIS, KHC |
| Consumer Discretionary | 10 | HD, MCD, NKE, SBUX, TGT, LOW, TJX, BKNG, CMG, DG |
| Industrials | 10 | CAT, DE, UNP, HON, GE, BA, LMT, RTX, MMM, UPS |
| Utilities | 10 | NEE, DUK, SO, D, AEP, SRE, EXC, XEL, ED, WEC |
| REITs | 9 | O, AMT, PLD, CCI, SPG, WELL, DLR, PSA, EQR (now VMRK) |
| Materials | 7 | LIN, APD, ECL, DD, NEM, FCX, NUE |
| Communication | 6 | DIS, CMCSA, NFLX, T, VZ, TMUS |

Daily prices come from the Yahoo Finance chart service, retrieved November 10, 2026, the same source the original paper used through the yfinance library. Each stock's history runs from its first available date through October 30, 2026. Sixteen stocks, including Johnson & Johnson, ExxonMobil, Coca-Cola, Procter & Gamble, Boeing, and Disney, start on January 2, 1962; the shortest history, Kraft Heinz, starts in July 2015. The median stock has 46.6 years and 11,750 trading days of data. The original paper did not say whether its prices included dividends. Here every return is a total return: RSI is computed on Yahoo's dividend-adjusted close, and the opening price is adjusted by the same factor, so trade returns and buy-and-hold returns both include reinvested dividends.

### 3.2 RSI Calculation

RSI uses Wilder's smoothing with a 14-period lookback: daily (or weekly) price changes are split into gains and losses, each is smoothed with an exponential average with $\alpha = 1/14$ after at least 14 periods, and RSI = 100 − 100 / (1 + average gain / average loss). For the weekly strategies, the dividend-adjusted close on the last trading day of each week is used before computing RSI.

### 3.3 Trading Rules and Execution

| Strategy | Timeframe | Buy signal | Sell signal |
|---|---|---|---|
| Daily 30/70 | Daily | RSI crosses below 30 | RSI crosses above 70 |
| Daily 20/80 | Daily | RSI crosses below 20 | RSI crosses above 80 |
| Weekly 30/70 | Weekly | RSI crosses below 30 | RSI crosses above 70 |
| Weekly 20/80 | Weekly | RSI crosses below 20 | RSI crosses above 80 |

A buy signal occurs when RSI is below the buy threshold after being at or above it on the prior bar; a sell signal occurs when RSI rises above the sell threshold after being at or below it. The original paper filled orders at the close of the signal bar, which a trader could not do exactly, because the signal is known only once that close prints. The rerun fills every order at the **next trading day's open**. Results under the original same-close rule are reported alongside and are nearly identical (Table 1). Positions are long-only, one per stock at a time, with no stop-loss. A position still open on October 30, 2026 is closed at that day's price and counted; Table 1 reports how many.

### 3.4 Benchmarks

The original paper compared trade returns with zero and the strategies with buying and holding each stock while idle cash earned nothing. The rerun adds four benchmarks.

1. **Random entries.** For every actual trade, a random entry day is drawn on the same stock and held for the same number of trading days, with the same next-open execution. Repeating this 1,000 times gives the distribution of average returns and win rates that a strategy with no timing ability would have earned on the same stocks with the same holding periods.
2. **The market over the same holding period.** For trades entered from January 1988, when Yahoo's S&P 500 total return index begins, each trade's return is compared with the index's return over the same dates.
3. **Buy-and-hold with Treasury bill interest.** The per-stock comparison is repeated with idle cash earning the 13-week Treasury bill rate.
4. **A portfolio.** Each strategy is run as an equal-weight portfolio across all stocks available on each day, with each stock's slice held in that stock while a position is open and in Treasury bills otherwise. It is compared with an equal-weight buy-and-hold portfolio of the same stocks, the S&P 500 total return index (from 1988), the S&P 500 price index (from 1962), and Treasury bills.

### 3.5 Transaction Costs

Results are gross unless stated. A 10 basis point round-trip cost is applied in Table 1 for reference. That is a modern figure and likely understates costs in the sample's early decades, when fixed public commission rates applied to U.S. exchange transactions; the SEC eliminated them after May 1975 under Rule 19b-3.

### 3.6 Statistical Inference

Trades on different stocks that are open over the same weeks share the same market moves, and RSI entries bunch up around market declines, so tests that treat trades as independent overstate the evidence. Tests on trade returns here use standard errors clustered two ways, by the month of entry and by stock (Cameron, Gelbach, & Miller, 2011); the conventional independent-trade t-statistic is shown for comparison. Stock-level comparisons with buy-and-hold use paired t-tests and Wilcoxon signed-

rank tests across the 99 stocks. Sector differences are tested with one-way ANOVA, as in the original paper, and with a clustered Wald test for the two 30/70 strategies; with four strategies tested, the Bonferroni-adjusted threshold is 0.05 / 4 = 0.0125.

SECTION IV

# 4. Results

## 4.1 Aggregate Performance

Table 1: Aggregate Strategy Performance (next-day open execution, total returns)

| Metric | Daily 30/70 | Daily 20/80 | Weekly 30/70 | Weekly 20/80 |
|---|---|---|---|---|
| Total trades | 3,948 | 599 | 600 | 83 |
| Stocks traded | 99 | 98 | 99 | 57 |
| Avg trades per stock | 39.9 | 6.1 | 6.1 | 0.8 |
| Win rate | 79.9% | 86.6% | 88.0% | 95.2% |
| Avg return per trade | 7.63% | 46.59% | 37.07% | 246.31% |
| Median return per trade | 9.06% | 27.56% | 34.37% | 169.30% |
| Std. dev. of returns | 15.37% | 88.59% | 38.64% | 256.20% |
| Avg holding period | 149 days | 769 days | 603 days | 2,381 days |
| Median holding period | 112 days | 525 days | 490 days | 2,093 days |
| Profit factor | 3.72 | 15.95 | 13.78 | 175.05 |
| Avg winning trade | +13.04% | +57.37% | +45.42% | +260.27% |

| | | | | |
|---|---|---|---|---|
| Avg losing trade | −13.96% | −23.33% | −24.17% | −29.37% |
| Worst trade | −93.79% | −87.64% | −87.47% | −65.73% |
| Best trade | +124.63% | +1247.84% | +215.41% | +1544.82% |
| Longest win streak | 69 | 53 | 83 | 51 |
| Longest losing streak | 9 | 3 | 6 | 2 |
| Positions open at end of data | 43 | 27 | 15 | 12 |
| Avg return after 10 bp round trip | 7.52% | 46.44% | 36.93% | 245.96% |
| Avg return, same-close execution | 7.69% | 47.59% | 36.94% | 242.50% |

Across the four strategies the rerun produced 5,230 trades. The patterns in the original paper all hold. Trade frequency falls sharply with stricter thresholds and longer timeframes, from about 40 trades per stock for Daily 30/70 to fewer than one for Weekly 20/80, which traded in only 57 of the 99 stocks. Win rates rise from 79.9% to 95.2%, and average returns per trade rise with the holding period, from 7.63% over about five months to 246% over about six and a half years. Executing at the next day's open rather than the signal-day close changes the average return by 1.0 percentage points or less for the first three strategies and by 3.8 points on Weekly 20/80's much larger average.

Table 1b: Original Paper Against the Rerun

| | Daily 30/70 | Daily 20/80 | Weekly 30/70 | Weekly 20/80 |
|---|---|---|---|---|
| Trades, original / rerun | 3,892 / 3,948 | 598 / 599 | 602 / 600 | 82 / 83 |
| Win rate, original / rerun | 79.4% / 79.9% | 87.3% / 86.6% | 87.7% / 88.0% | 96.3% / 95.2% |
| Avg return, original / rerun | 7.66% / 7.63% | 47.37% / 46.59% | 37.09% / 37.07% | 244.10% / 246.31% |

The rerun matches the original paper's trade counts within 2% and its win rates and average returns closely, which indicates the original backtest was run on real data much as described; the small differences come from the next-open execution, the total-return prices, the extra eight months of data, and the loss of AvalonBay.

### 4.2 Does RSI Timing Beat Random Timing?

Table 2: Are the Trades Better Than Zero, and Better Than Random Entries?

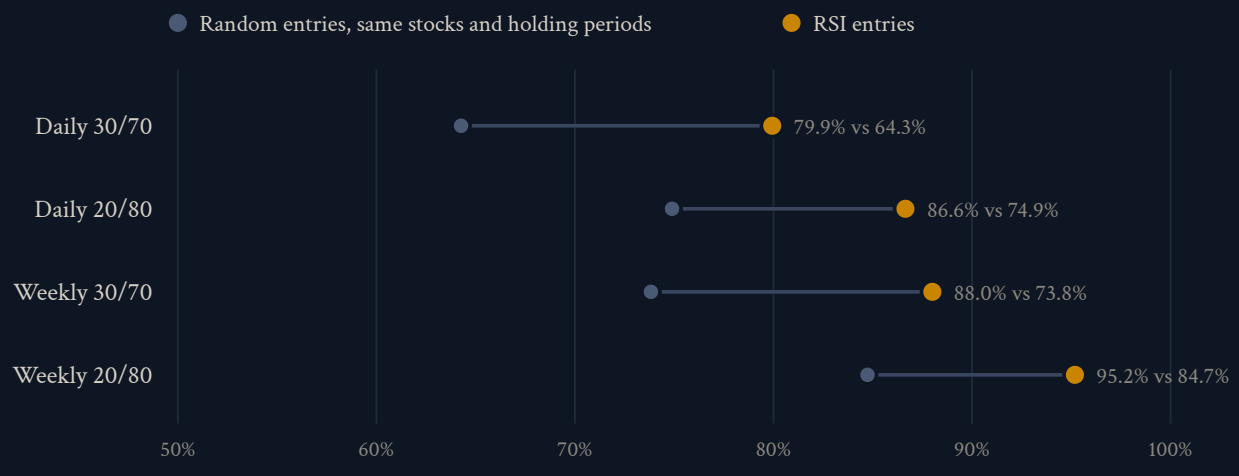
| Strategy | Avg return | Naive t | Clustered t | p (clustered) | Random-entry avg | Random 90% range | p vs random | Win rate | R w R |
|---|---|---|---|---|---|---|---|---|---|
| Daily 30/70 | 7.63% | 31.17 | 19.95 | $1.4 \times 10^{-88}$ | 6.96% | 6.41% to 7.54% | 0.032 | 79.9% | 64 |
| Daily 20/80 | 46.59% | 12.87 | 9.86 | $6.0 \times 10^{-23}$ | 46.98% | 39.60% to 55.66% | 0.490 | 86.6% | 74 |
| Weekly 30/70 | 37.07% | 23.50 | 10.10 | $5.7 \times 10^{-24}$ | 31.35% | 27.34% to 35.43% | 0.010 | 88.0% | 73 |
| Weekly 20/80 | 246.31% | 8.76 | 9.43 | $4.0 \times 10^{-21}$ | 195.22% | 143.88% to 256.22% | 0.081 | 95.2% | 84 |

Every strategy's average trade is positive, and with standard errors clustered by entry month and stock the t-statistics fall (for Daily 30/70, from 31.17 to 19.95) but remain very large. That only shows the trades made money on stocks that rose over time. The random-entry benchmark asks the sharper question: would random entries on the same stocks, held for the same lengths of time, have done as well?

On average return, the answer is nearly. Random entries matched to the Daily 30/70 trades averaged 6.96% against the strategy's 7.63%, a difference of about 0.7 percentage points per trade that random draws exceed only 3% of the time. Weekly 30/70 shows a larger edge (37.07% against 31.35%, p = 0.010), Weekly 20/80 a large but statistically marginal one (p = 0.081), and Daily 20/80 none (46.59% against 46.98%). Because random draws treat entries as independent while real RSI entries cluster in time, these p-values somewhat overstate the evidence.

On win rate and median return, the RSI trades are far ahead of random entries (Figure 1). Much of that gap is built into the rules rather than earned by the entry signal: a position closes only when RSI climbs above 70 or 80, which happens after a run of gains, so trades end at local highs by construction, while a random window of the same length ends wherever it happens to end. The exit rule lifts the win rate and the median without lifting the average by as much, which is what Table 2 shows.

FIGURE 1. WIN RATE, RSI ENTRIES AGAINST RANDOM ENTRIES




Random entries are drawn on the same stocks and held for the same number of trading days as each actual trade, 1,000 times over. RSI trades end after a rally by design, which accounts for much of the gap.

Table 3: Trade Returns Against the S&P 500 Total Return Over the Same Dates

| Strategy | Trades from 1988 | Avg trade return | S&P 500 TR, same dates | Avg excess | Clustered t | p-value | Share beating index |
|---|---|---|---|---|---|---|---|
| Daily 30/70 | 2,933 | 8.15% | 6.24% | 1.91% | 4.89 | $9.9 \times 10^{-7}$ | 63.2% |
| Daily 20/80 | 388 | 51.97% | 38.23% | 13.74% | 2.81 | 0.0050 | 61.6% |
| Weekly 30/70 | 450 | 38.20% | 27.98% | 10.22% | 3.96 | $7.6 \times 10^{-5}$ | 64.7% |
| Weekly 20/80 | 58 | 279.82% | 170.36% | 109.46% | 3.38 | 0.0007 | 82.8% |

Against the S&P 500 total return over the same dates, the average trade from 1988 on beat the index for every strategy, by 1.91% per trade for Daily 30/70 (clustered t = 4.89). That comparison flatters the strategies, however, because the stocks themselves beat the index by a wide margin over the period: an equal-weight portfolio of the same stocks, simply held, returned 18.1% a year from 1988 against the index's 11.4% (Table 5). Excess returns over the index measure the advantage of owning these survivors at least as much as the advantage of RSI timing; the random-entry test, which holds the stocks fixed, is the cleaner measure of timing.

## 4.3 Per-Stock Comparison With Buy-and-Hold

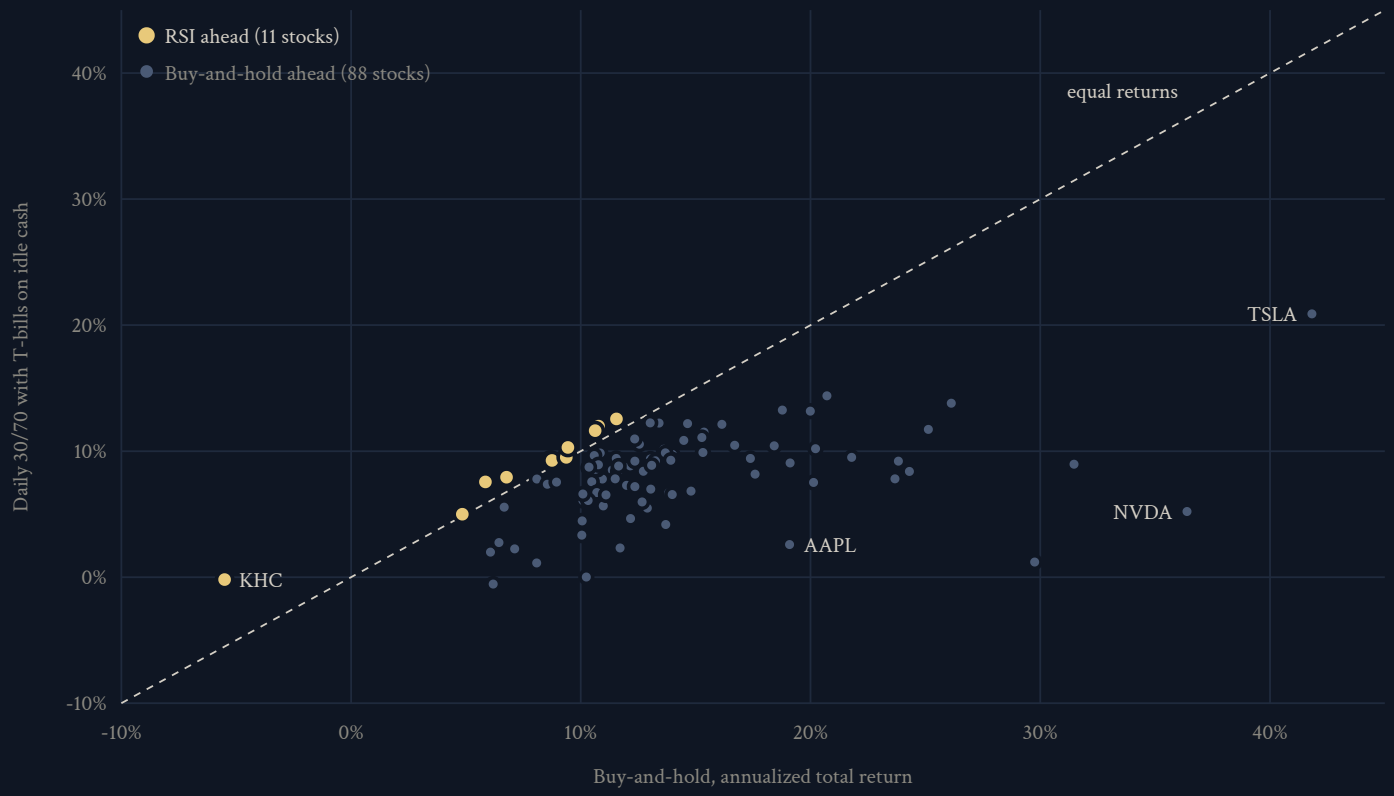
Table 4: Per-Stock Annualized Returns Against Buying and Holding the Same Stock (averages across 99 stocks)

| Strategy | Time invested | Annual return, cash at 0% | Annual return, cash in T-bills | Buy-and-hold | Paired t (0%) | Paired t (T-bills) | p (T-bills) | Wilcoxon p (T-bills) | Stocks where RSI wins (T-bills) |
|---|---|---|---|---|---|---|---|---|---|

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Daily 30/70 | 38% | 5.9% | 8.2% | 13.7% | −13.80 | −9.39 | $2.5 \times 10^{-15}$ | $5.2 \times 10^{-16}$ | 11 of 99 |
| Daily 20/80 | 28% | 4.1% | 6.7% | 13.7% | −15.77 | −11.26 | $2.3 \times 10^{-19}$ | $4.2 \times 10^{-17}$ | 6 of 99 |
| Weekly 30/70 | 23% | 4.0% | 6.8% | 13.7% | −17.83 | −12.19 | $2.4 \times 10^{-21}$ | $1.1 \times 10^{-17}$ | 3 of 99 |
| Weekly 20/80 | 12% | 2.1% | 5.2% | 13.7% | −16.87 | −12.06 | $4.5 \times 10^{-21}$ | $4.3 \times 10^{-17}$ | 7 of 99 |

Stock by stock, every strategy trailed buying and holding the same stock. With idle cash at 0%, as in the original paper, the Daily 30/70 rule returned 5.9% a year on average against 13.7% for buy-and-hold. Giving idle cash the Treasury bill rate narrows the gap but does not close it: 8.2% against 13.7%, with the strategy ahead in only 11 of 99 stocks (paired t = −9.39). The other strategies trail by more, and the Wilcoxon tests agree for all four.

FIGURE 2. PER-STOCK ANNUALIZED RETURN, DAILY 30/70 AGAINST BUY-AND-HOLD




Each dot is one stock, over its full history. The strategy earns Treasury bill interest while out of the market. Points below the dashed line are stocks where holding beat the strategy.

The shortfall is largest for the stocks that rose most. Tesla, Nvidia, and Amazon returned more than 30% a year held, and RSI captured only a fraction of that; the 11 stocks where RSI came out ahead, such as Kraft Heinz, UPS, and Schlumberger, were all slow growers that returned 12% a year or less held.

## 4.4 Portfolio-Level Results

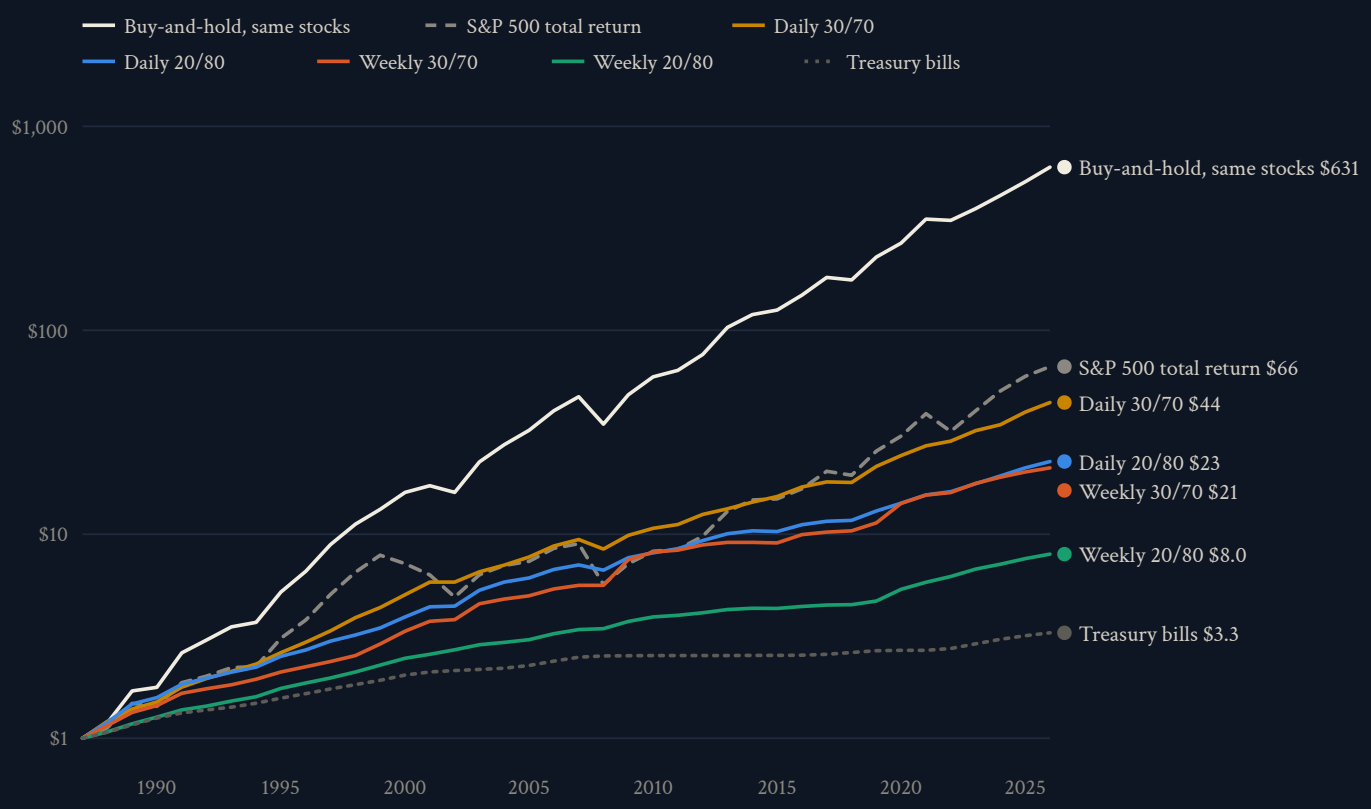
Table 5: Portfolio Results, January 1988 to October 2026

| Portfolio | Annual return | Volatility | Sharpe ratio | Max drawdown | $1 grew to | Avg invested |
|---|---|---|---|---|---|---|
| Daily 30/70 | 10.3% | 11.1% | 0.67 | −32.3% | $44.3 | 37% |

| | | | | | | |
|---|---|---|---|---|---|---|
| Daily 20/80 | 8.4% | 6.6% | 0.80 | −17.7% | $22.8 | 29% |
| Weekly 30/70 | 8.2% | 8.0% | 0.65 | −21.2% | $21.2 | 22% |
| Weekly 20/80 | 5.5% | 3.1% | 0.79 | −6.7% | $8.0 | 11% |
| Buy-and-hold, same stocks (equal weight) | 18.1% | 16.8% | 0.90 | −45.0% | $631 | |
| S&P 500 total return | 11.4% | 17.5% | 0.51 | −55.3% | $66.4 | |
| Treasury bills | 3.1% | 0.1% | n/a | −0.0% | $3.3 | |

Run as portfolios, the strategies look different from the per-stock view. From January 1988 the Daily 30/70 portfolio returned 10.3% a year with 11.1% volatility and a worst drawdown of −32.3%, against 11.4%, 17.5%, and −55.3% for the S&P 500 total return index. All four RSI portfolios had higher Sharpe ratios than the index and much smaller drawdowns, because they spent most of their time in Treasury bills and were invested mainly after sharp declines. All four also earned less than the index in absolute terms, and all four trailed the equal-weight buy-and-hold portfolio of the same stocks on return (18.1%) and on Sharpe ratio (0.90).

FIGURE 3. GROWTH OF $1, JANUARY 1988 TO OCTOBER 2026




Log scale. RSI portfolios hold each stock's slice in Treasury bills while out of a position. The buy-and-hold line holds the same 99 stocks in equal weight; its lead over the index reflects survivorship and hindsight in the stock list as much as anything else.

## Table 5b: Portfolio Results, March 1962 to October 2026

| Portfolio | Annual return | Volatility | Sharpe ratio | Max drawdown | $1 grew to | Avg invested |
|---|---|---|---|---|---|---|
| Daily 30/70 | 10.2% | 10.4% | 0.56 | −32.3% | $524 | 38% |
| Daily 20/80 | 8.6% | 6.8% | 0.60 | −25.9% | $208 | 31% |
| Weekly 30/70 | 8.6% | 7.3% | 0.56 | −25.7% | $202 | 24% |
| Weekly 20/80 | 6.5% | 3.4% | 0.57 | −18.2% | $57.4 | 15% |

| | | | | | |
|---|---|---|---|---|---|
| Buy-and-hold, same stocks (equal weight) | 16.2% | 16.0% | 0.75 | −48.4% | $16,862 |
| S&P 500 price index (no dividends) | 7.5% | 16.2% | 0.24 | −56.8% | $108 |
| Treasury bills | 4.7% | 0.2% | n/a | −0.0% | $19.2 |

Over the full period from 1962 the RSI portfolios again trailed the same stocks held, on return (6.5% to 10.2% a year against 16.2%) and on Sharpe ratio (0.56 to 0.60 against 0.75). The S&P 500 price index omits dividends, so it is not a like-for-like benchmark before 1988 and is shown only for reference. The fair comparison throughout is with the same stocks held, because both the RSI portfolios and the buy-and-hold portfolio are built from the same survivor list: against that benchmark the RSI rules lost on return and on risk-adjusted return in both periods.

### 4.5 Sector Analysis

Table 6: Performance by Sector, Daily 30/70

| Sector | Trades | Win rate | Avg return | Median return |
|---|---|---|---|---|
| Mega Cap Tech | 169 | 78.1% | 10.54% | 12.34% |
| Consumer Discretionary | 373 | 78.8% | 8.80% | 10.79% |
| Materials | 290 | 80.7% | 8.29% | 10.00% |
| Energy | 383 | 78.9% | 8.18% | 10.74% |
| Healthcare | 442 | 80.8% | 8.01% | 9.40% |
| Consumer Staples | 418 | 82.3% | 7.87% | 8.02% |
| REITs | 247 | 80.2% | 7.61% | 8.60% |

| | | | | |
|---|---|---|---|---|
| Financials | 391 | 78.0% | 7.45% | 9.53% |
| Industrials | 551 | 77.3% | 6.77% | 8.93% |
| Utilities | 465 | 86.0% | 6.25% | 7.53% |
| Communication | 219 | 74.9% | 5.69% | 7.70% |

Table 7: Performance by Sector, Weekly 30/70

| Sector | Trades | Win rate | Avg return | Median return |
|---|---|---|---|---|
| Mega Cap Tech | 29 | 89.7% | 62.48% | 60.46% |
| Consumer Discretionary | 53 | 83.0% | 47.17% | 36.72% |
| Financials | 64 | 90.6% | 44.01% | 44.78% |
| Materials | 51 | 92.2% | 43.17% | 43.63% |
| REITs | 34 | 82.4% | 37.49% | 31.97% |
| Communication | 38 | 84.2% | 34.81% | 32.91% |
| Energy | 66 | 89.4% | 33.48% | 28.04% |
| Industrials | 94 | 88.3% | 33.21% | 33.16% |
| Consumer Staples | 51 | 86.3% | 32.08% | 34.21% |
| Healthcare | 59 | 89.8% | 30.92% | 30.01% |
| Utilities | 61 | 88.5% | 24.93% | 24.00% |

Table 8: Tests of Equal Sector Means

| Strategy | ANOVA F | ANOVA p | Clustered Wald p | Significant? |
|---|---|---|---|---|
| Daily 30/70 | 1.866 | 0.045 | 0.021 | At 0.05 only; not after Bonferroni (0.0125) |
| Daily 20/80 | 0.604 | 0.811 | 0.770 | No |
| Weekly 30/70 | 3.051 | 0.0009 | 0.0014 | Yes; also after Bonferroni (0.0125) |
| Weekly 20/80 | 0.884 | 0.552 | not reliable* | No |

* With only 2 to 15 Weekly 20/80 trades per sector, the clustered Wald test is not reliable and is not used.

Mega Cap Tech again tops both 30/70 tables, and Utilities again has the highest Daily 30/70 win rate (86.0%) with one of the lowest average returns. As in the original paper, sector means differ significantly for Weekly 30/70 even after a Bonferroni correction, but for Daily 30/70 only at the 5% level, and not at all for the 20/80 strategies. The Mega Cap Tech ranking carries hindsight: its seven names were chosen because they were mega-caps in 2026.

### 4.6 Time Periods and Crises

Table 9: Daily 30/70 by Decade of Entry (all trades)

| Period | Trades | Win rate | Avg return | Avg excess over S&P 500 TR |
|---|---|---|---|---|
| 1962 to 1969 | 156 | 74.4% | 4.56% | n/a (index starts 1988) |
| 1970 to 1979 | 348 | 72.1% | 3.88% | n/a (index starts 1988) |
| 1980 to 1989 | 586 | 82.4% | 8.36% | 2.97% |
| 1990 to 1999 | 630 | 81.3% | 9.99% | 1.54% |
| 2000 to 2009 | 657 | 76.7% | 6.42% | 6.23% |

| | | | | |
|---|---|---|---|---|
| 2010 to 2019 | 871 | 84.7% | 8.07% | 0.73% |
| 2020 to 2026 | 700 | 78.9% | 8.00% | –0.43% |

The original paper broke out only 2005 to 2025. Every decade from the 1960s on produced positive average Daily 30/70 trades, with the weakest results in the 1960s and 1970s and the strongest in the 1980s and 1990s. Against the S&P 500 total return over the same dates, the excess return has faded: it was largest in 2000 to 2009 (6.23% per trade), small in 2010 to 2019 (0.73%), and slightly negative since 2020 (–0.43%).

Table 10: Trades Entered During Market Crises

| Window (entry dates) | Strategy | Trades | Win rate | Avg return | Avg excess over S&P 500 TR |
|---|---|---|---|---|---|
| GFC (Oct 2007 to Mar 2009) | Daily 30/70 | 113 | 54.9% | –1.31% | 7.47% |
| | Daily 20/80 | 20 | 85.0% | 72.04% | 24.93% |
| | Weekly 30/70 | 72 | 86.1% | 40.22% | 14.18% |
| | Weekly 20/80 | 11 | 100.0% | 352.35% | 33.68% |
| COVID (Feb to Jun 2020) | Daily 30/70 | 64 | 70.3% | 9.68% | –3.27% |
| | Daily 20/80 | 22 | 95.5% | 67.47% | 19.55% |
| | Weekly 30/70 | 47 | 95.7% | 74.75% | 14.76% |
| | Weekly 20/80 | 18 | 100.0% | 239.38% | 90.66% |
| 2022 bear market | Daily 30/70 | 118 | 68.6% | 6.01% | –0.38% |

| | | | | | |
|---|---|---|---|---|---|
| | Daily 20/80 | 21 | 90.5% | 36.09% | −9.83% |
| | Weekly 30/70 | 24 | 83.3% | 27.96% | −5.06% |
| | Weekly 20/80 | 2 | 100.0% | 217.48% | 187.01% |

The crisis results match the original paper. Daily 30/70 trades entered during the financial crisis lost money on average (−1.31%, with a 54.9% win rate), because the rule kept buying a market that kept falling, while the stricter and weekly rules, which waited for deeper oversold readings, did very well; all 11 Weekly 20/80 trades from the crisis were winners, averaging 352%. In COVID-era entries every strategy except Daily 30/70 beat the index over the same dates. These windows contain few trades, entered within weeks of each other, in stocks known to have survived, so they describe how survivors bought near crisis lows performed, not how the signal would fare on stocks that failed.

## 4.7 Individual Stocks

Table 11: Top 10 Stocks by Average Return, Daily 30/70

| Rank | Ticker | Sector | Trades | Win rate | Avg return |
|---|---|---|---|---|---|
| 1 | TSLA | Mega Cap Tech | 17 | 88.2% | 21.81% |
| 2 | AMZN | Mega Cap Tech | 24 | 83.3% | 15.19% |
| 3 | AMT | REITs | 23 | 91.3% | 13.73% |
| 4 | AMGN | Healthcare | 41 | 85.4% | 12.54% |
| 5 | TJX | Consumer Discretionary | 36 | 86.1% | 12.51% |
| 6 | EOG | Energy | 33 | 78.8% | 12.42% |
| 7 | LIN | Materials | 27 | 92.6% | 12.32% |

| | | | | | |
|---|---|---|---|---|---|
| 8 | ABBV | Healthcare | 16 | 87.5% | 12.15% |
| 9 | SBUX | Consumer Discretionary | 31 | 90.3% | 12.07% |
| 10 | MSFT | Mega Cap Tech | 34 | 82.4% | 11.81% |

Table 12: Bottom 10 Stocks by Average Return, Daily 30/70

| Rank | Ticker | Sector | Trades | Win rate | Avg return |
|---|---|---|---|---|---|
| 90 | DG | Consumer Discretionary | 14 | 71.4% | 4.61% |
| 91 | FCX | Materials | 30 | 73.3% | 4.58% |
| 92 | ED | Utilities | 62 | 87.1% | 4.40% |
| 93 | MRK | Healthcare | 50 | 72.0% | 4.23% |
| 94 | EQR | REITs | 28 | 75.0% | 3.90% |
| 95 | HAL | Energy | 47 | 68.1% | 3.85% |
| 96 | C | Financials | 47 | 66.0% | 3.56% |
| 97 | AAPL | Mega Cap Tech | 37 | 70.3% | 2.23% |
| 98 | KHC | Consumer Staples | 11 | 54.5% | 0.34% |
| 99 | TMUS | Communication | 14 | 50.0% | 0.13% |

The best and worst names are largely the same as in the original paper. Tesla, Amazon, American Tower, Amgen, and TJX lead; T-Mobile, Kraft Heinz, Apple, and Citigroup trail. Apple's place near the bottom illustrates the central result: its average RSI trade earned only 2.23%, because the rule sold after every rally and sat out much of the stock's long climb, while holding Apple returned about 19% a year from 1980.

SECTION V

# 5. Discussion

## 5.1 Winning Trades, Losing Strategy

Every strategy wins most of its trades yet trails buy-and-hold, and the rerun now measures why. The Daily 30/70 rule was invested only 38% of the time on average, Weekly 20/80 only 12%. While invested the strategies earned roughly 20% to 21% a year (for example, 7.63% over an average of 149 days for Daily 30/70 annualizes to 19.7%), but for the rest of the time the money sat in Treasury bills while the stocks kept rising. Earning about 20% a year for a third of the time and the bill rate otherwise cannot keep up with stocks that compounded at 13.7% a year on average.

## 5.2 What the Signal Adds

The random-entry test (Table 2) gives the clearest answer the data allow. The entry signal adds a modest edge for the 30/70 rules, about 0.7 percentage points per daily trade and about 6 per weekly trade, and nothing measurable for Daily 20/80. The very high win rates owe much to the exit rule, which only sells into strength. Buying weakness in stocks that kept growing paid off, but mostly because the stocks kept growing.

## 5.3 Daily vs. Weekly Timeframes

Daily RSI behaves like a short-to-medium-term mean-reversion system: about 40 trades per stock, average returns near 7.6%, and holding periods of about five months, with about one trade in five losing money. Weekly RSI behaves like a rare, opportunistic system: about six trades per stock for 30/70 and fewer than one for 20/80, with large returns over holding periods of years, many of them entered in major declines (Table 10).

### 5.4 Threshold Selection: 30/70 vs. 20/80

Stricter thresholds raise win rates and returns per trade but cut the number of signals sharply and lengthen holding periods. They also lower time in the market, which is why the 20/80 portfolios have the lowest volatility and drawdowns in Table 5 and the lowest returns. The Daily 20/80 rule shows no entry edge over random timing at all.

### 5.5 Survivorship Bias

The universe was chosen in 2026 from companies that were then large and listed, so firms that failed, were acquired, or shrank out of contention during 1962 to 2026 are absent by construction (Brown, Goetzmann, Ibbotson, & Ross, 1992). The rerun can now show the size of the effect on this list: holding the 99 stocks in equal weight returned 18.1% a year from 1988, against 11.4% for the S&P 500 total return index, and 16.2% a year from 1962. Part of that gap is the equal weighting, but much of it is hindsight in the choice of names. Any strategy that holds these stocks inherits that tailwind, which is why the RSI trades beat the index over the same dates (Table 3) and why the absolute results should be read as an upper bound. The comparisons that hold the stock list fixed, against random entries and against buying and holding the same stocks, are largely immune to this bias, and both point the same way. AvalonBay's disappearance from the data after its 2026 merger is a reminder that the list itself erodes: a point-in-time database that includes delisted firms, such as CRSP, would be needed to remove the bias.

### 5.6 Transaction Costs and Taxes

For Daily 30/70, a 10 basis point round-trip cost lowers the average trade from 7.63% to 7.52%. Most Daily 30/70 trades last less than a year, so gains are taxed at short-term rates. At a 37% rate, taxing winning trades only ($79.9\% \times 13.04\% \times 37\% = 3.86\%$) gives about 3.66% per trade after costs and taxes, a lower bound; if losses fully offset gains, the same rate applied to the after-cost average gives about

4.74%. Buy-and-hold defers taxes and qualifies for long-term rates, so costs and taxes widen the gap in Table 4.

### 5.7 Data Snooping

The four parameter sets are industry standards published since 1978, not values optimized on this data. Sullivan, Timmermann, and White (1999) applied White's Reality Check (White, 2000) to 7,846 technical rules on the Dow Jones Industrial Average and found that the best rule's in-sample significance did not carry over to their 1987 to 1996 out-of-sample period. This study applies no Reality Check, but its main conclusion does not depend on picking the best rule: none of the four beats holding the same stocks.

### 5.8 Limitations

The universe is survivor-selected (Section 5.5). Yahoo's early-decade data and its dividend adjustments are not audited. The random-entry benchmark draws entries uniformly over each stock's history, so it does not match the calendar timing of RSI entries, and its draws are independent while real entries cluster. The portfolio backtest rebalances to equal weight every day without cost, holds idle cash at the 13-week bill rate, and assumes trades fill at the open with no slippage. The 10 basis point cost reflects modern markets. The rules are long-only, with no stop-loss or position sizing.

SECTION VI

## 6. Practical Implications

This section summarizes what the hypothetical backtest shows. It describes findings, not recommendations, and the caveats in Sections 5.5 and 5.8 apply throughout.

### 6.1 Standalone Rules Versus Buy-and-Hold

As standalone, long-only rules, none of the four strategies matched holding the same stocks, stock by stock or as a portfolio, with or without interest on idle cash. Time out of the market outweighed positive returns per trade.

### 6.2 Risk

Because they sit in Treasury bills most of the time, the RSI portfolios had much lower volatility and drawdowns than the index or buy-and-hold, and higher Sharpe ratios than the index. That smoother ride came with lower returns, and against holding the same stocks it did not translate into a better Sharpe ratio. Individual trades still carried large losses: the worst Daily 30/70 trade lost −93.8%.

### 6.3 Where RSI Helped Most

The entry signal's edge over random timing was clearest for the weekly 30/70 rule and in deep market declines, when stricter rules bought near crisis lows. Whether RSI adds value as one input to a broader process, such as timing purchases an investor already plans to make, was not tested here.

SECTION VII

## 7. Conclusions

This study backtests four RSI strategies on real daily total-return prices for 99 U.S. stocks over up to 64 years, 5,230 trades in all. Our principal findings are:

1. The rerun reproduces the original results. Daily 30/70 made 3,948 trades with a 79.9% win rate and a 7.63% average return; Weekly 20/80 made 83 trades with a 95.2% win rate and a 246% average return.

2. The trades beat random timing only modestly. Against random entries on the same stocks held for the same periods, average returns were higher for Daily 30/70 ($p = 0.032$) and Weekly 30/70 ($p = 0.010$), marginally for Weekly 20/80, and not at all for Daily 20/80. The high win rates come largely from an exit rule that sells only after rallies.
3. Every strategy trailed buying and holding the same stocks: stock by stock, with or without Treasury bill interest on idle cash, and as portfolios since 1962 and since 1988.
4. As portfolios, the strategies had lower volatility, smaller drawdowns, and higher Sharpe ratios than the S&P 500, but lower returns than the index.
5. Sector, decade, and crisis results match the original paper; the excess over the index has faded since 2010.
6. The stock list is survivor-selected, and holding it beat the index by about 7 percentage points a year since 1988, so absolute results are an upper bound; the comparisons that hold the list fixed are the reliable ones.

Buying RSI oversold readings and selling overbought readings made money on these stocks, mostly because the stocks rose; the signal itself added a modest edge for the 30/70 rules and little else, and sitting out of the market between signals cost more than the timing earned.

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**Revision note, November 2026.** This version reruns the backtest on real Yahoo Finance prices retrieved November 10, 2026, through October 30, 2026. It uses total-return prices and next-day-open execution, reports positions still open at the end, adds a random-entry benchmark, trade returns against the S&P 500 total return index, a buy-and-hold comparison with Treasury bill interest on idle cash, an equal-weight portfolio backtest with Sharpe ratios and drawdowns, clustered standard errors, and a full decade-by-decade breakdown, and it adds three figures. AvalonBay Communities is dropped because its price history is no longer available after its 2026 merger. The October 2026 revision had corrected statistical and descriptive errors and listed this analysis as still needed; the rerun confirms the original paper's trade-level results and replaces the open questions with the results above. Sections 1 and 2 are unchanged apart from minor edits.

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# Appendix: Data and Software

Prices: Yahoo Finance chart service, daily open, close, and dividend-adjusted close, retrieved November 10, 2026, through October 30, 2026. Treasury bill rate: Yahoo's 13-week Treasury bill yield (^IRX). Benchmarks: S&P 500 total return index (^SP500TR, from January 1988) and S&P 500 price index (^GSPC). Analysis in Python with pandas, NumPy, SciPy, and statsmodels. The random-entry benchmark uses 1,000 draws with a fixed seed. Data and analysis code are available from the author on request.



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